CoreMP135

SKU:K135

描述

CoreMP135 是一款基于 **STM32MP135DAE7** 芯片的一体化 **Linux** 工控主机。其集成了单核 ARM Cortex-A7 处理器，主频高达 1 GHz ，另配备 4Gbit DDR3L SDRAM 运行内存。 适用于高级工业自动化、智能家居和多媒体娱乐设备以及工业物联网边缘网关，机器人运动控制中枢等领域。

CoreMP135 提供多个功能丰富的接口：

- 2 路千兆网 GbE 接口
- 1 路高清视频输出接口
- 2 路 USB2.0-A 接口
- 1 路 USB-C 接口 (支持 OTG 及供电) ， microSD 卡槽
- 2 路 CAN FD 接口
- 1 路 PWR485 (9~24V 电源输入 + RS485) 接口
- 2 个 Grove (I2C & UART) 接口

在人机交互界面方面，**CoreMP135** 自带 2.0 寸 IPS 电容触摸屏，以及 1W 扬声器 (16 bits I2S 驱动) 。整体低功耗设计，采用 AXP2101 电源管理芯片，内置 RTC (BM8563) 定时唤醒 - 休眠功能，支持充电电池供电，配有 DC 电源插座，支持外部 DC 12V@2A 直流电源供电。随主机配备 microSD 卡，其内预装 Debian 系统，开机即用；适配多种安装场景，设备底部配备 DIN 导轨底板，方便挂墙以及螺丝固定。

教程 & 快速上手

UiFlow2

本教程将向你介绍，如何通过 UiFlow2 图形化编程平台控制 CoreMP135 设备。

产品特性

- STM32MP135DAE7@Arm Cortex-A7@1GHz
- Linux 标准平台
- 多种通讯方式，丰富的外设接口 (CANRS485 千兆网口等)
- 2.0 寸触摸屏幕
- 统一电源管理
- 内置扬声器

- microSD 以及 4Gbit DDR3L SDRAM 运行内存
- M5-Bus&PORT A/C
- DIN Rail 导轨方便安装
- 开发平台
    - UiFlow2

包装内容

- 1 x CoreMP135
- 1 x M3 六角扳手
- 1 x HT3.96-4P
- 2 x 接线端子 2.54mm-2P (绿色)
- 1 x microSD 卡 (已经装于机器中)
- 1 x 说明书

应用场景

- 工业自动化
- 智能家居
- 工业物联网边缘网关
- 教育和开发
- 机器人运动中枢控制器

规格参数

规格	参数
MCU	STM32MP135DAE7@单核 Arm Cortex-A7 处理器，主频 1 GHz
电源管理芯片	AXP2101
485 通讯	MAX3485
CAN 通讯	两路 SIT1051T/3 (高速 FDCAN)
USB 集线器接口	GL852G (2x USB2.0) 1x USBC (支持 OTG 及供电)
高清视频接口芯片	LT8618SXB，支持最高 24 位的色深
DDR3L SDRAM	4Gbit
以太网	RTL8211F（最高支持 1Gbps 的数据速率） 2x RJ45
RTC 时钟	BM8563
屏幕	ILI9342C (2.0 IPS LCD) 分辨率：240 x 320px
屏幕触摸	FT6336U
功放	NS4168 (单声道 D 类功放) I2S 串行数字音频输入 支持宽范围采样速率：8kHz~96kHz
扬声器	2014 型腔体喇叭：1W@8Ω
直流电源输入	DC 12V@2A
工作温度	0 ~ 40°C
供电电源	DC 12V@2A 或者 USB-C 5V@3A
产品尺寸	81.9 x 54.0 x 39.5mm
产品重量	98.3g
包装尺寸	122.2 x 71.5 x 61.6mm
毛重	154.7g

操作说明

M5-Bus 电源

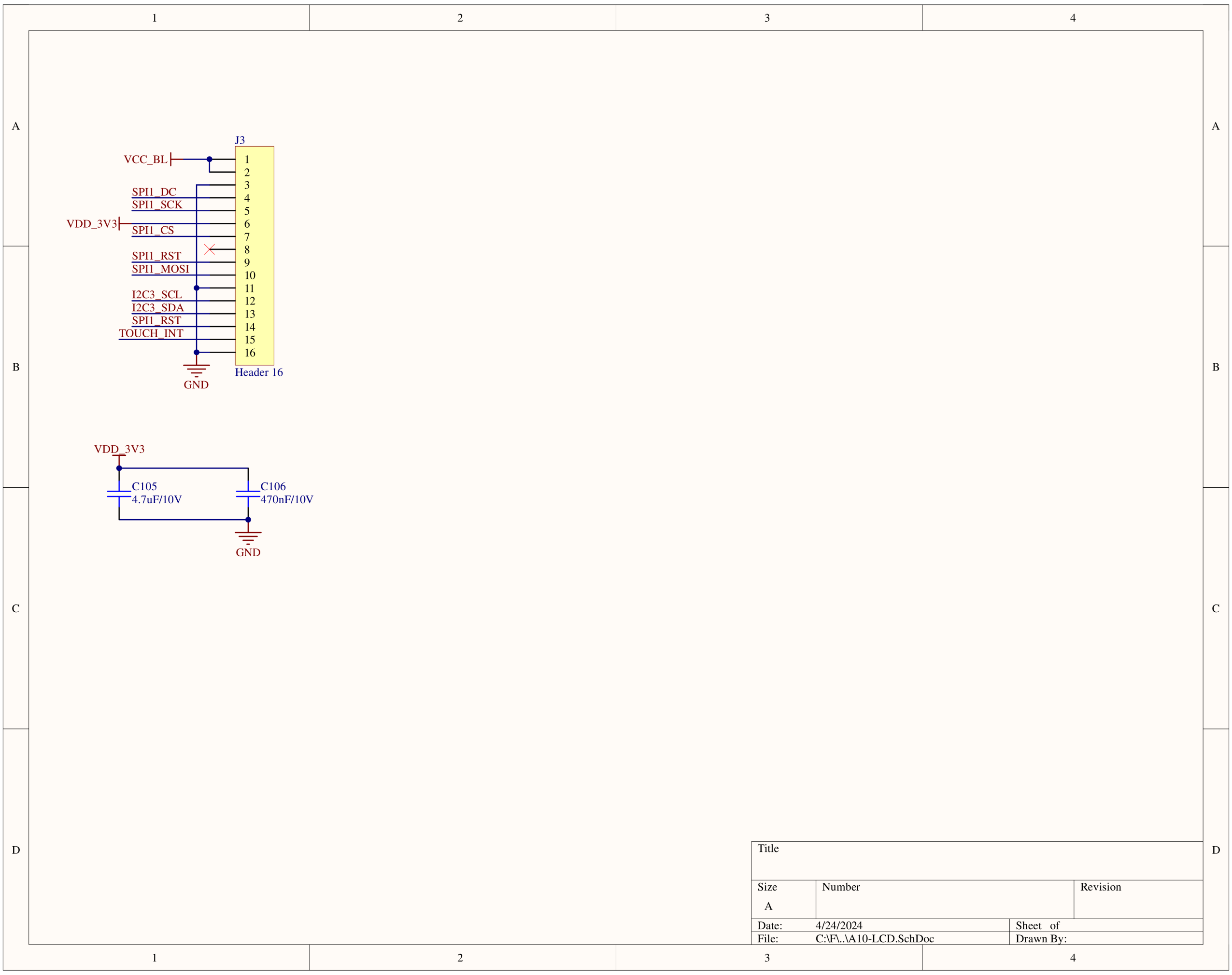
CoreMP135 的 M5-Bus 电源总线输入输出控制：

参考原理图将 BUS_OUT_EN 设置为低电平则总线 5V 为输入模式，高电平则总线 5V 为输出模式。可使用以下指令打开向下输出：

```
echo 131 > /sys/class/gpio/export && echo out > /sys/class/gpio/PI3/direction
echo 1 > /sys/class/gpio/PI3/value
# echo 0 > /sys/class/gpio/PI3/value
```

原理图

- CoreMP135 原理图 PDF
- CoreMP135 MidLayer 原理图 PDF


J3
VCC_BL
SPI1_DC
SPI1_SCK
VDD_3V3
SPI1_CS
SPI1_RST
SPI1_MOSI
I2C3_SCL
I2C3_SDA
SPI1_RST
TOUCH_INT
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
GND
Header 16
VDD_3V3
C105
4.7uF/10V
C106
470nF/10V
GND
Title
Size
A
Number
Revision
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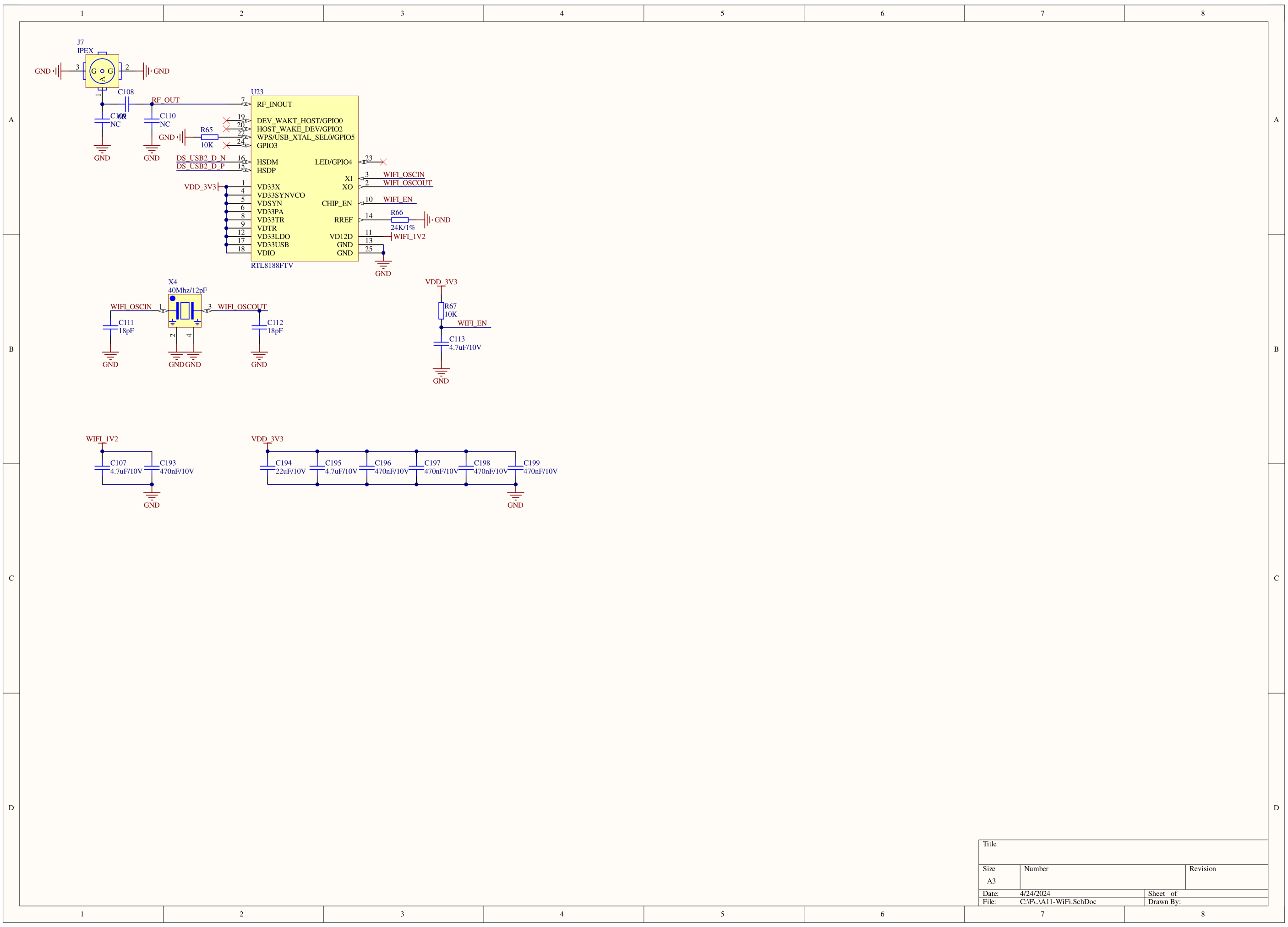
J7
IPEX
GND
GND
C108
RF_OUT
C110
NC
GND
GND
U23
RF_INOUT
DEV_WAKT_HOST/GPIO0
HOST_WAKE_DEV/GPIO2
WPS/USB_XTAL_SEL0/GPIO5
GPIO3
R65
10K
HSDM
HSDP
LED/GPIO4
DS_USB2_D_N
DS_USB2_D_P
XI
XO
WIFI_OSCIN
WIFI_OSCOUT
VDD_3V3
VD33X
VD33SYNVCO
VDSYN
VD33PA
VD33TR
VDTR
VD33LDO
VD33USB
VDIO
CHIP_EN
WIFI_EN
RREF
R66
24K/1%
VD12D
WIFI_1V2
GND
RTL8188FTV
X4
40Mhz/12pF
WIFI_OSCIN
WIFI_OSCOUT
C111
18pF
C112
18pF
GND
VDD_3V3
R67
10K
WIFI_EN
C113
4.7uF/10V
WIFI_1V2
C107
4.7uF/10V
C193
470nF/10V
VDD_3V3
C194
22uF/10V
C195
4.7uF/10V
C196
470nF/10V
C197
470nF/10V
C198
470nF/10V
C199
470nF/10V
Title
Size
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Number
Revision
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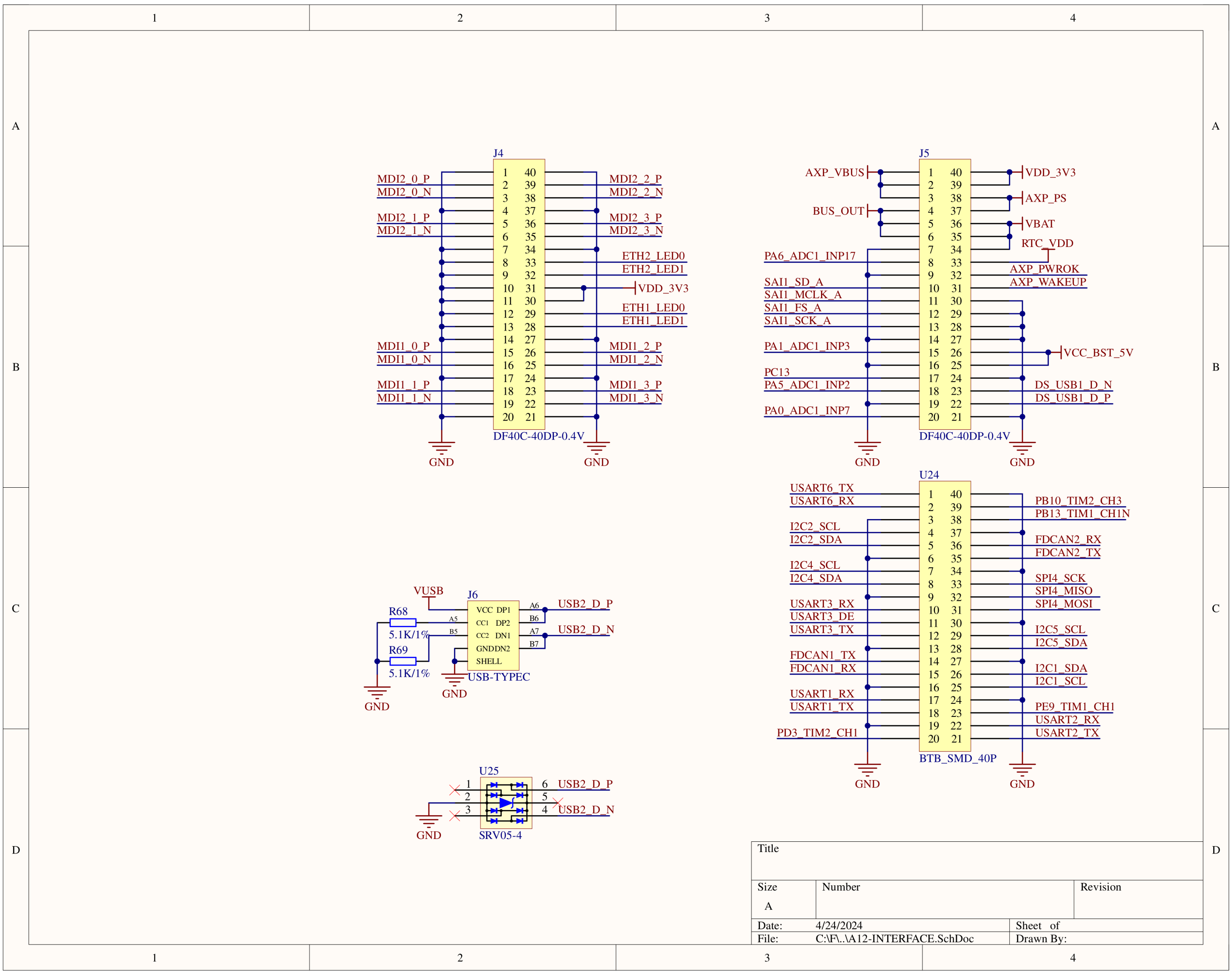
J4
MDI2_0_P
MDI2_0_N
MDI2_1_P
MDI2_1_N
MDI1_0_P
MDI1_0_N
MDI1_1_P
MDI1_1_N
MDI2_2_P
MDI2_2_N
MDI2_3_P
MDI2_3_N
ETH2_LED0
ETH2_LED1
VDD_3V3
ETH1_LED0
ETH1_LED1
MDI1_2_P
MDI1_2_N
MDI1_3_P
MDI1_3_N
DF40C-40DP-0.4V
GND
J5
AXP_VBUS
BUS_OUT
PA6_ADC1_INP17
SAI1_SD_A
SAI1_MCLK_A
SAI1_FS_A
SAI1_SCK_A
PA1_ADC1_INP3
PC13
PA5_ADC1_INP2
PA0_ADC1_INP7
VDD_3V3
AXP_PS
VBAT
RTC_VDD
AXP_PWROK
AXP_WAKEUP
VCC_BST_5V
DS_USB1_D_N
DS_USB1_D_P
DF40C-40DP-0.4V
U24
USART6_TX
USART6_RX
I2C2_SCL
I2C2_SDA
I2C4_SCL
I2C4_SDA
USART3_RX
USART3_DE
USART3_TX
FDCAN1_TX
FDCAN1_RX
USART1_RX
USART1_TX
PD3_TIM2_CH1
PB10_TIM2_CH3
PB13_TIM1_CH1N
FDCAN2_RX
FDCAN2_TX
SPI4_SCK
SPI4_MISO
SPI4_MOSI
I2C5_SCL
I2C5_SDA
I2C1_SDA
I2C1_SCL
PE9_TIM1_CH1
USART2_RX
USART2_TX
BTB_SMD_40P
VUSB
J6
R68
5.1K/1%
R69
5.1K/1%
USB2_D_P
USB2_D_N
USB-TYPEC
U25
SRV05-4
Title
Size
A
Number
Revision
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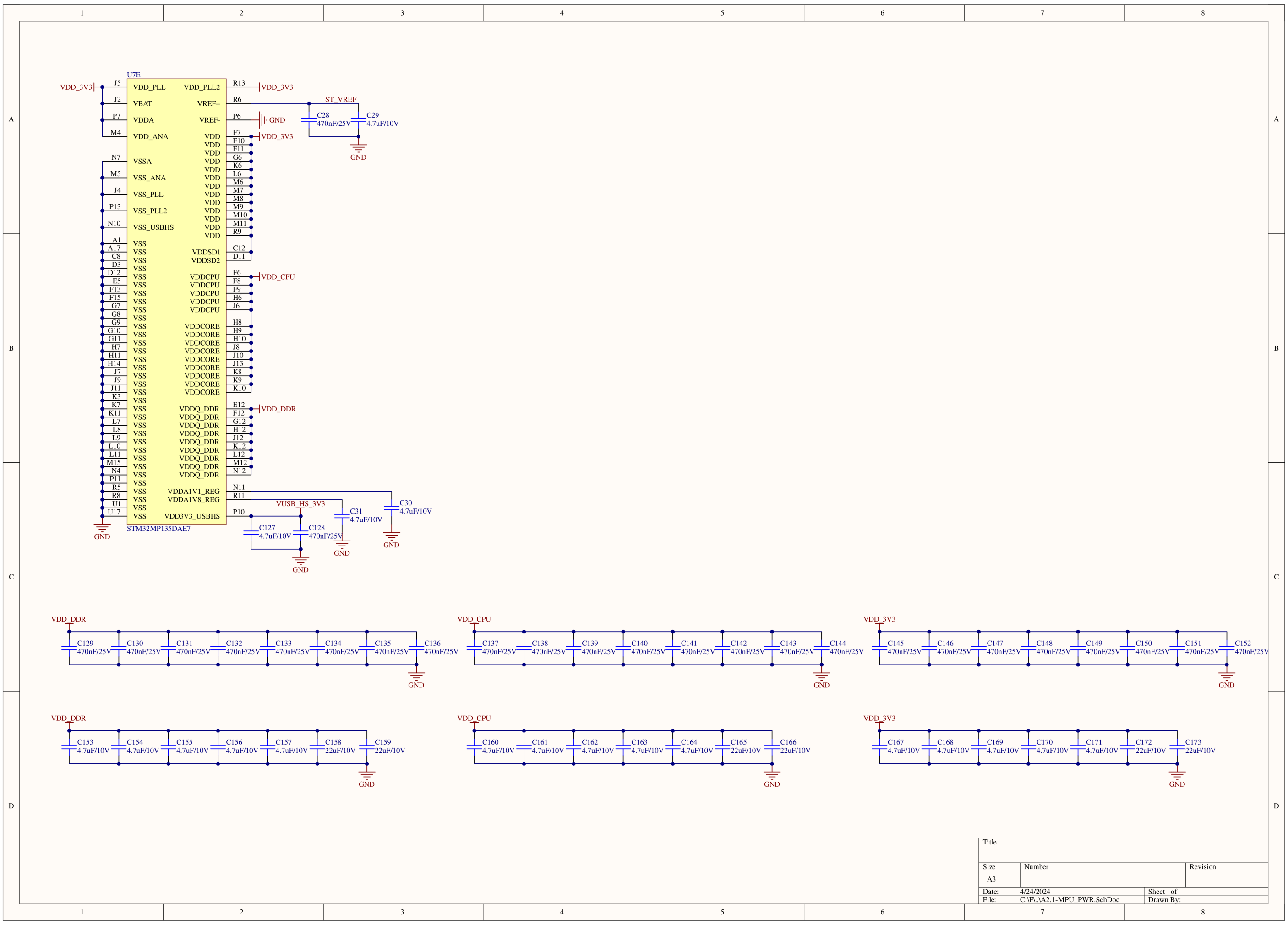

U7E
STM32MP135DAE7
VDD_3V3
VDD_CPU
VDD_DDR
VUSB_HS_3V3
ST_VREF
GND
C28 470nF/25V
C29 4.7uF/10V
C30 4.7uF/10V
C31 4.7uF/10V
C127 4.7uF/10V
C128 470nF/25V
Title
Size A3
Number
Revision
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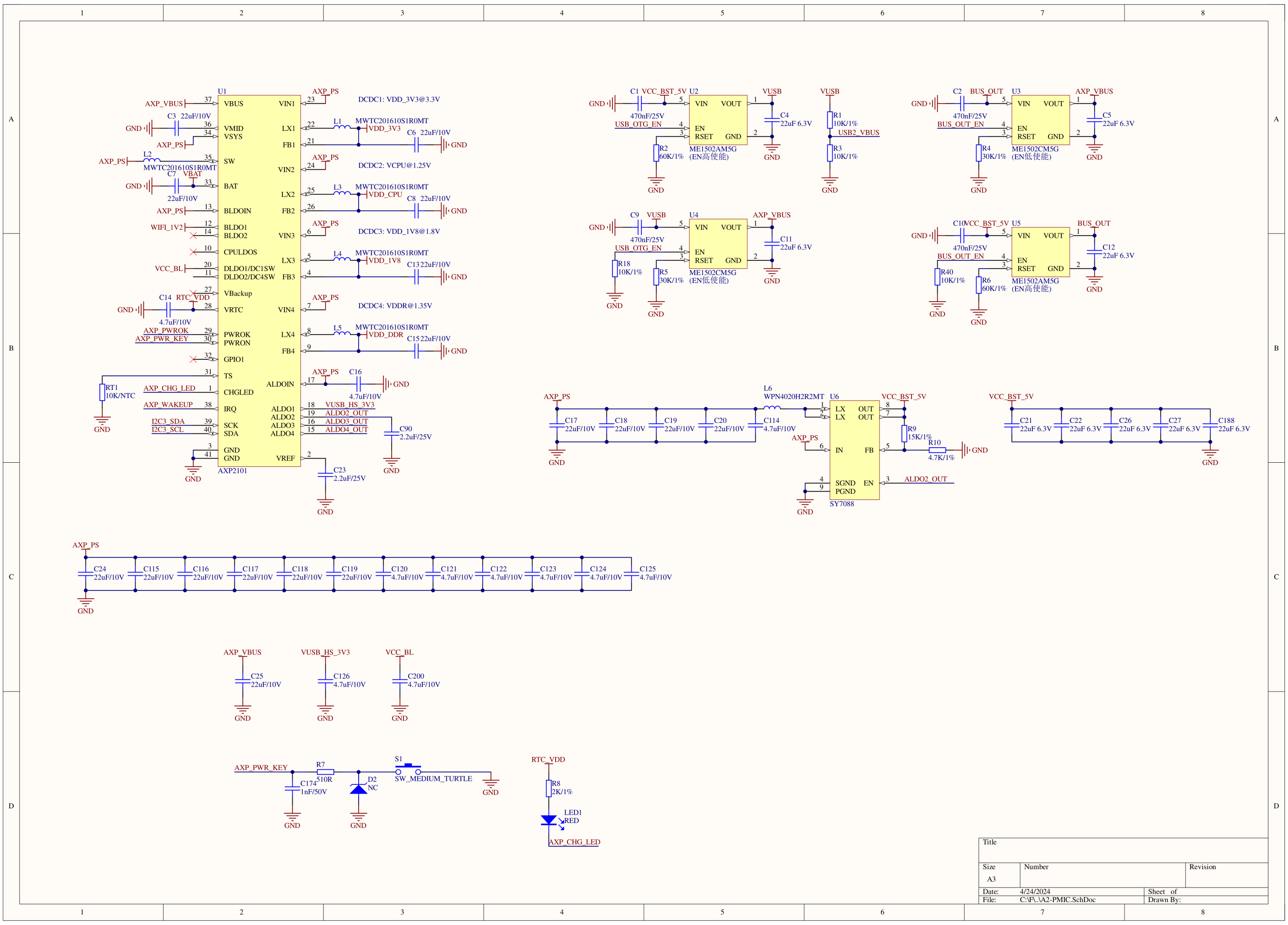
DCDC1: VDD_3V3@3.3V
DCDC2: VCPU@1.25V
DCDC3: VDD_1V8@1.8V
DCDC4: VDDR@1.35V
AXP2101
ME1502AM5G
(EN高使能)
ME1502CM5G
(EN低使能)
SY7088
SW_MEDIUM_TURTLE
Title
Size
A3
Number
Revision
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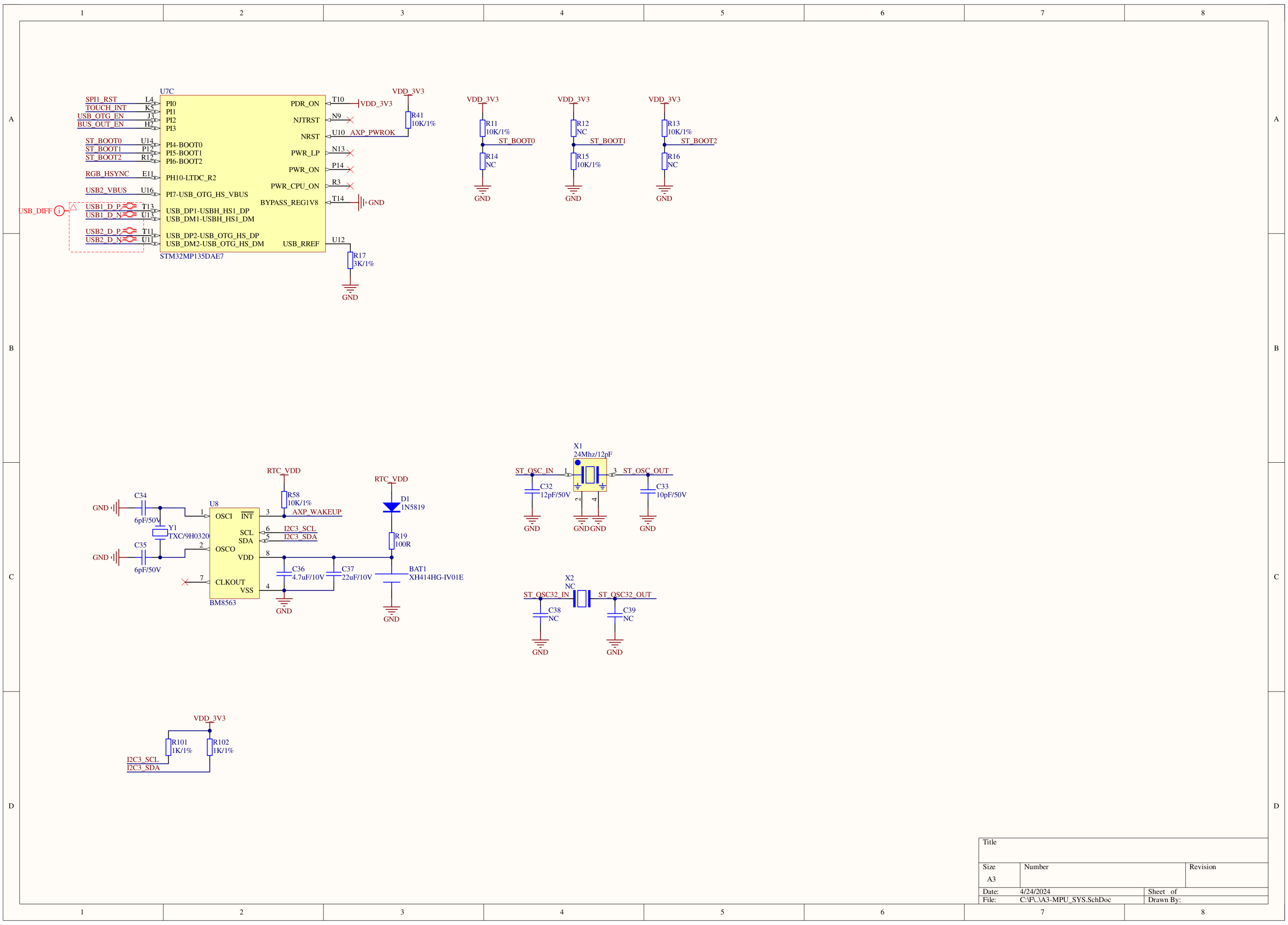
U7C
STM32MP135DAE7
SPI1_RST
TOUCH_INT
USB_OTG_EN
BUS_OUT_EN
ST_BOOT0
ST_BOOT1
ST_BOOT2
RGB_HSYNC
USB2_VBUS
USB1_D_P
USB1_D_N
USB2_D_P
USB2_D_N
USB_DIFF
PDR_ON
NJTRST
NRST
PWR_LP
PWR_ON
PWR_CPU_ON
BYPASS_REG1V8
USB_RREF
AXP_PWROK
VDD_3V3
R41 10K/1%
R17 5K/1%
GND
R11 10K/1%
R14 NC
R12 NC
R15 10K/1%
R13 10K/1%
R16 NC
X1 24Mhz/12pF
ST_OSC_IN
ST_OSC_OUT
C32 12pF/50V
C33 10pF/50V
X2 NC
ST_OSC32_IN
ST_OSC32_OUT
C38 NC
C39 NC
U8 BM8563
RTC_VDD
R58 10K/1%
AXP_WAKEUP
I2C3_SCL
I2C3_SDA
C34 6pF/50V
C35 6pF/50V
Y1 TXC/9H0320
C36 4.7uF/10V
C37 22uF/10V
D1 1N5819
R19 100R
BAT1 XH414HG-IV01E
R101 1K/1%
R102 1K/1%
Title
Size A3
Number
Revision
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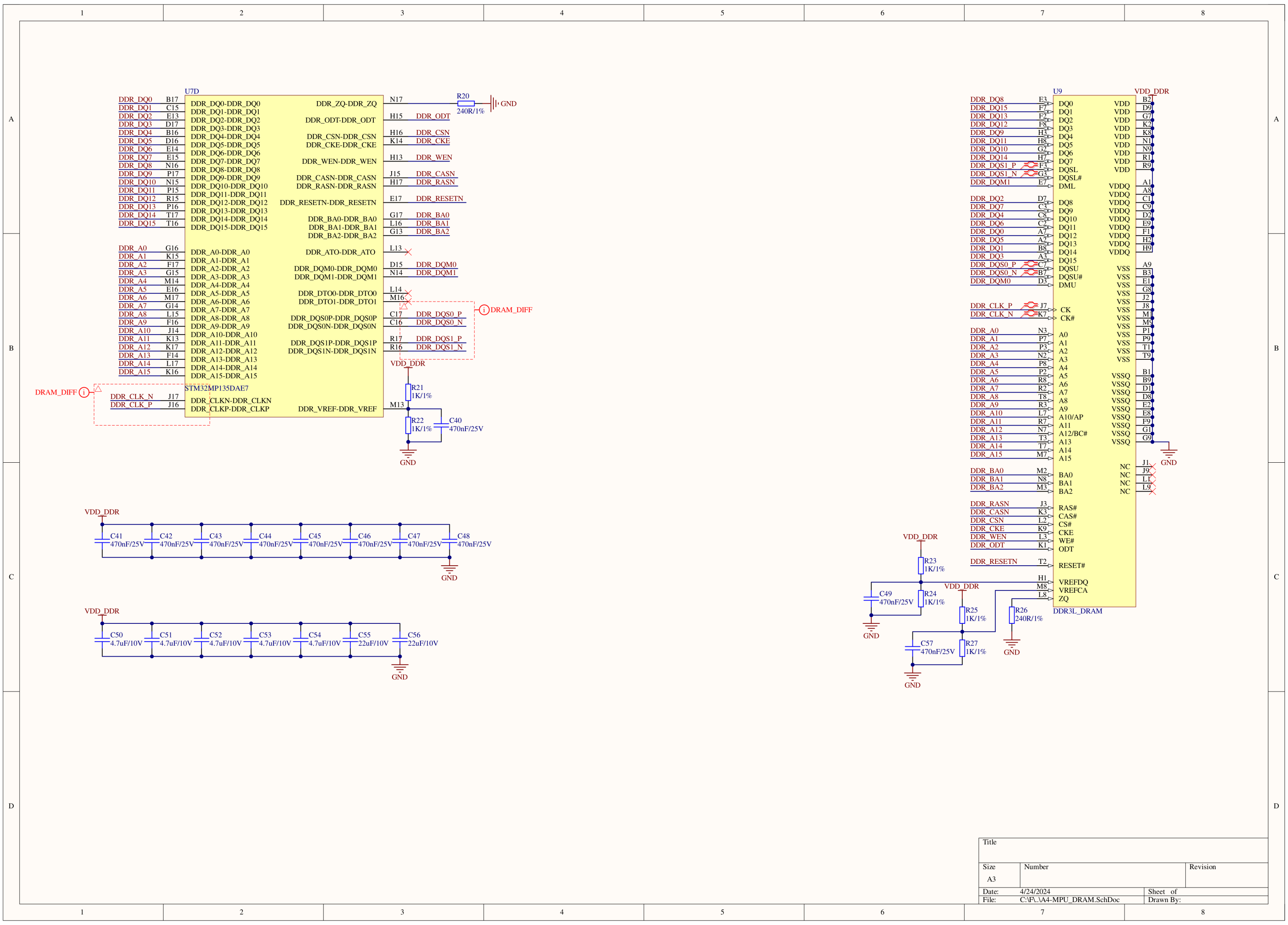
U7D
STM32MP135DAE7
U9
DDR3L_DRAM
VDD_DDR
GND
DRAM_DIFF
R20 240R/1%
R21 1K/1%
R22 1K/1%
C40 470nF/25V
R23 1K/1%
R24 1K/1%
R25 1K/1%
R26 240R/1%
R27 1K/1%
C49 470nF/25V
C57 470nF/25V
C41 470nF/25V
C42 470nF/25V
C43 470nF/25V
C44 470nF/25V
C45 470nF/25V
C46 470nF/25V
C47 470nF/25V
C48 470nF/25V
C50 4.7uF/10V
C51 4.7uF/10V
C52 4.7uF/10V
C53 4.7uF/10V
C54 4.7uF/10V
C55 22uF/10V
C56 22uF/10V
Title
Size
A3
Number
Revision
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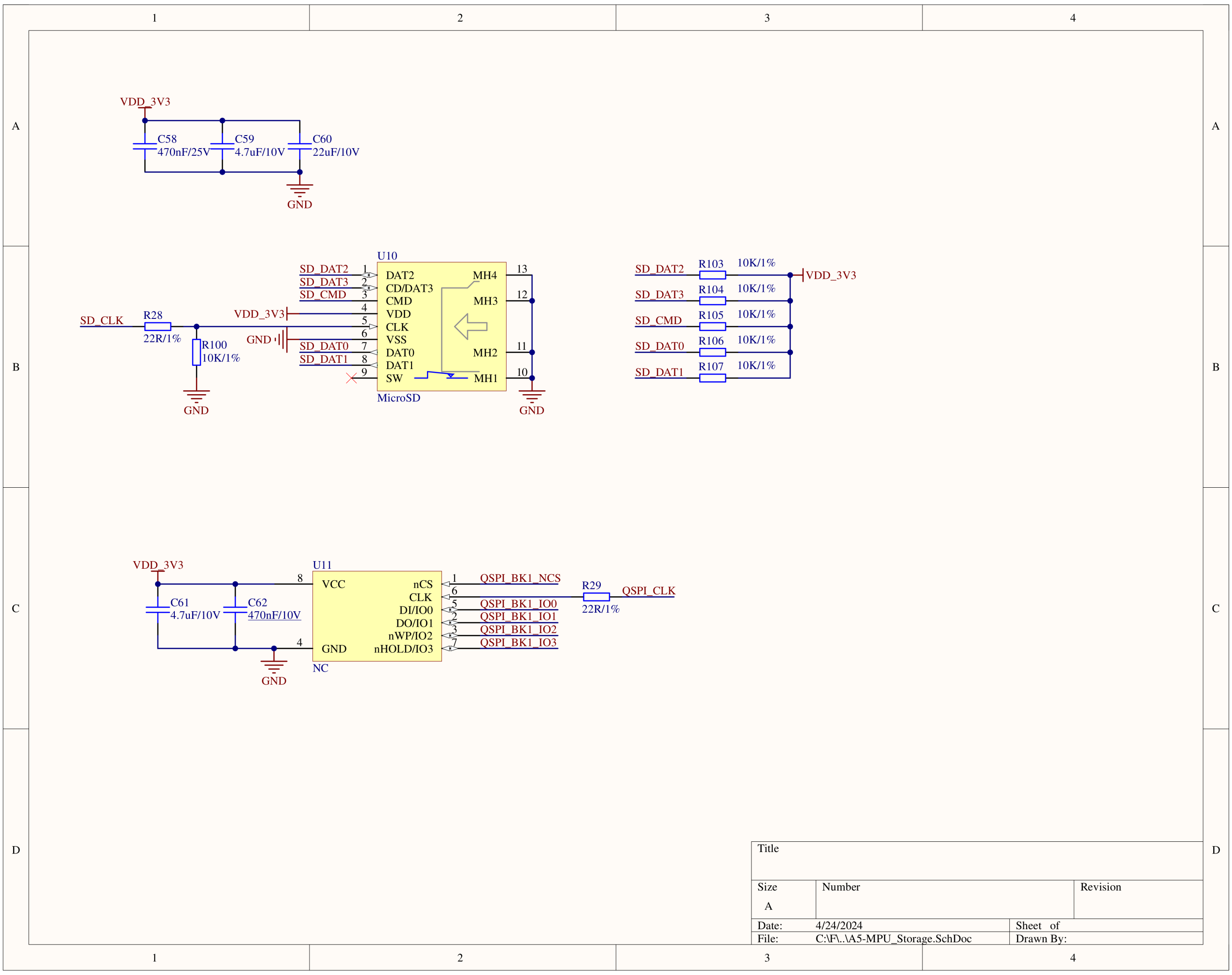
VDD_3V3
C58
470nF/25V
C59
4.7uF/10V
C60
22uF/10V
GND
U10
SD_DAT2
SD_DAT3
SD_CMD
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
SW
MH4
MH3
MH2
MH1
MicroSD
SD_CLK
R28
22R/1%
R100
10K/1%
SD_DAT0
SD_DAT1
R103
R104
R105
R106
R107
10K/1%
U11
VCC
GND
nCS
CLK
DI/IO0
DO/IO1
nWP/IO2
nHOLD/IO3
NC
C61
4.7uF/10V
C62
470nF/10V
QSPI_BK1_NCS
QSPI_BK1_IO0
QSPI_BK1_IO1
QSPI_BK1_IO2
QSPI_BK1_IO3
R29
22R/1%
QSPI_CLK
Title
Size
A
Number
Revision
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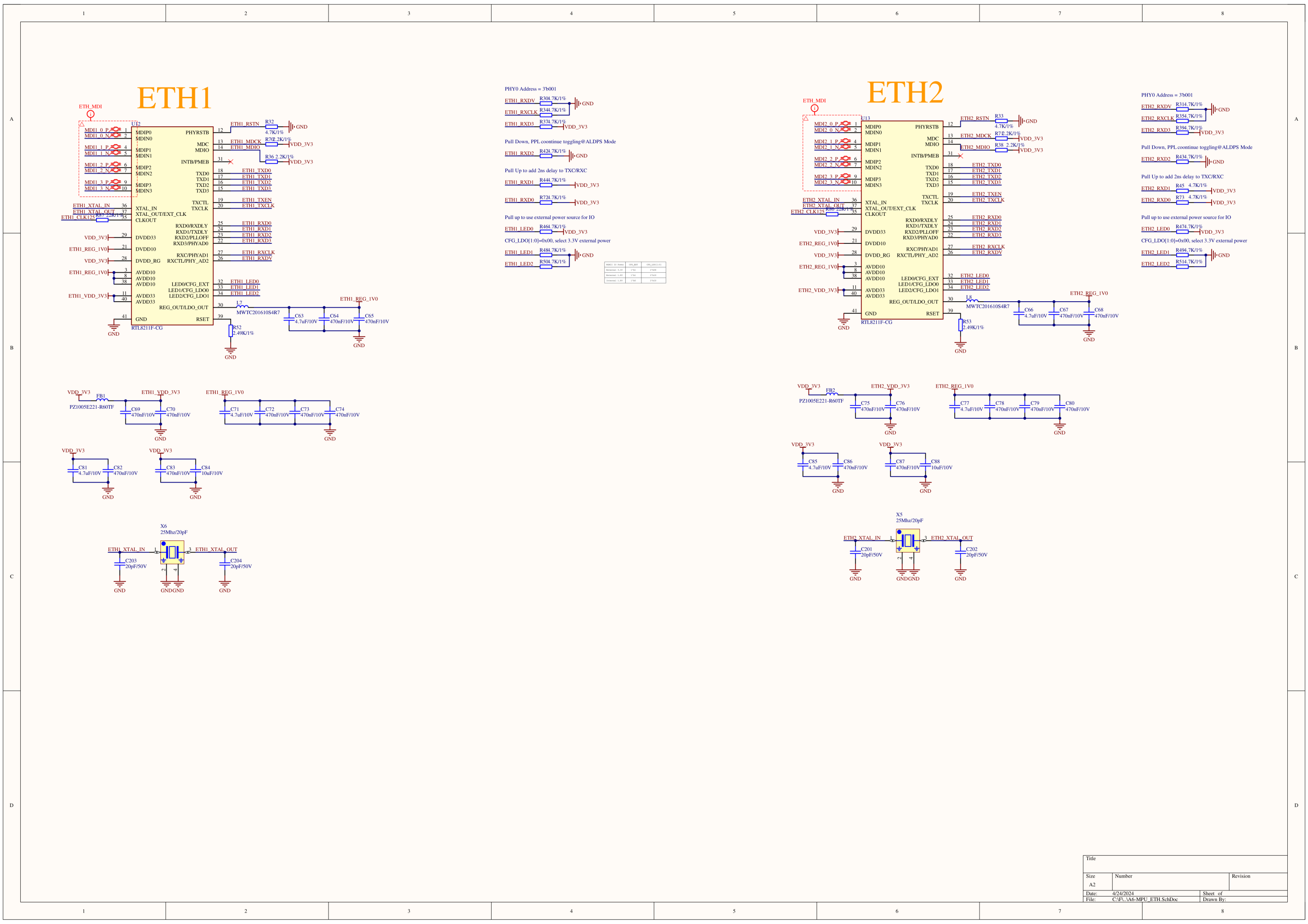
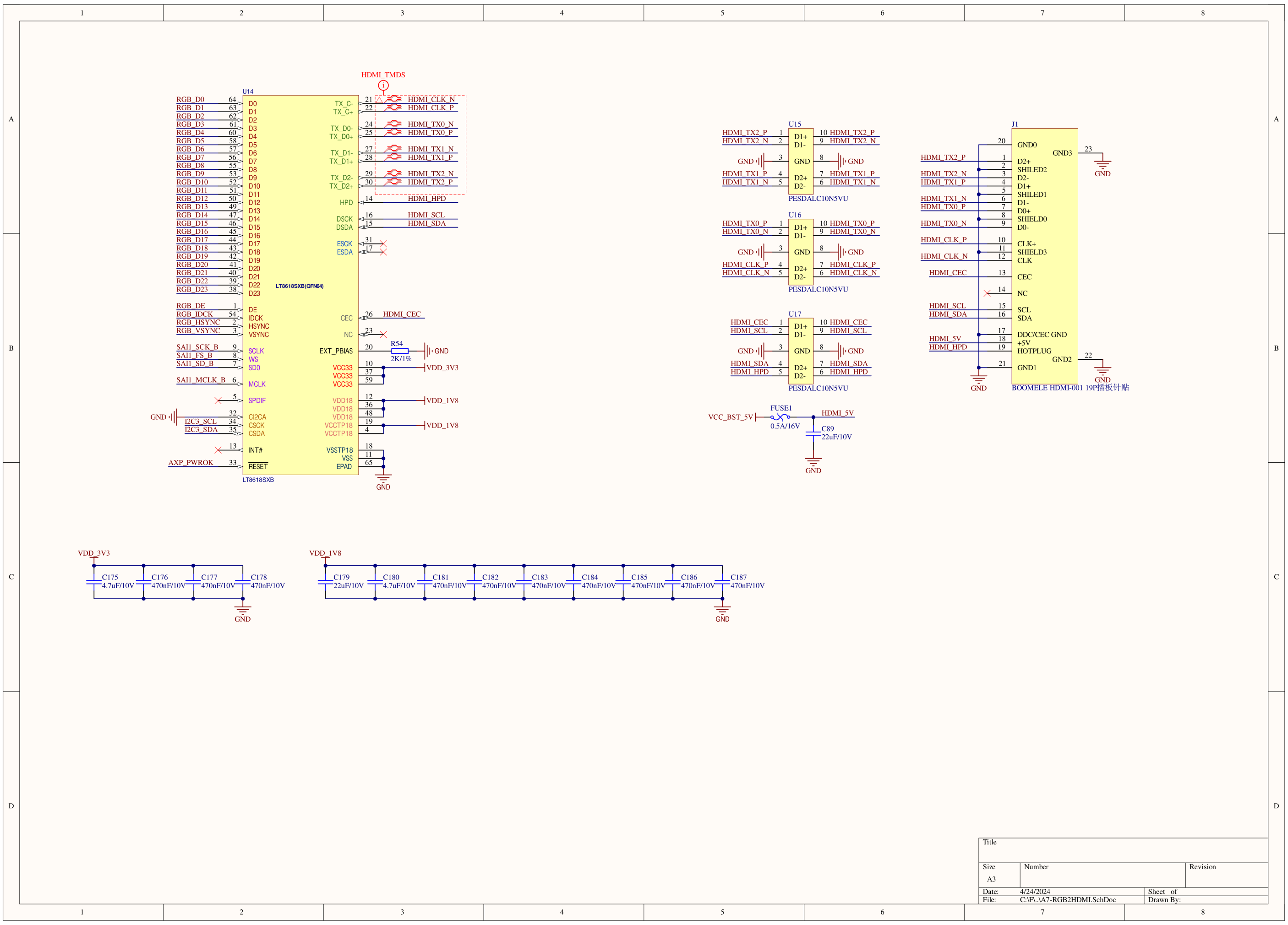

ETH1
ETH2
PHY0 Address = 3'b001
Pull Down, PPL coontinue toggling@ALDPS Mode
Pull Up to add 2ns delay to TXC/RXC
Pull up to use external power source for IO
CFG_LDO[1:0]=0x00, select 3.3V external power
RTL8211F-CG
25Mhz/20pF


HDMI_TMDS
U14
LT8618SXB(QFN64)
LT8618SXB
U15
PESDALC10N5VU
U16
PESDALC10N5VU
U17
PESDALC10N5VU
J1
BOOMELE HDMI-001 19P插板针贴
FUSE1
0.5A/16V
VCC_BST_5V
HDMI_5V
C89
22uF/10V
R54
2K/1%
VDD_3V3
VDD_1V8
GND
C175
4.7uF/10V
C176
470nF/10V
C177
470nF/10V
C178
470nF/10V
C179
22uF/10V
C180
4.7uF/10V
C181
470nF/10V
C182
470nF/10V
C183
470nF/10V
C184
470nF/10V
C185
470nF/10V
C186
470nF/10V
C187
470nF/10V
Title
Size
A3
Number
Revision
Date:
4/24/2024
File:
C:\F\..\A7-RGB2HDMI.SchDoc
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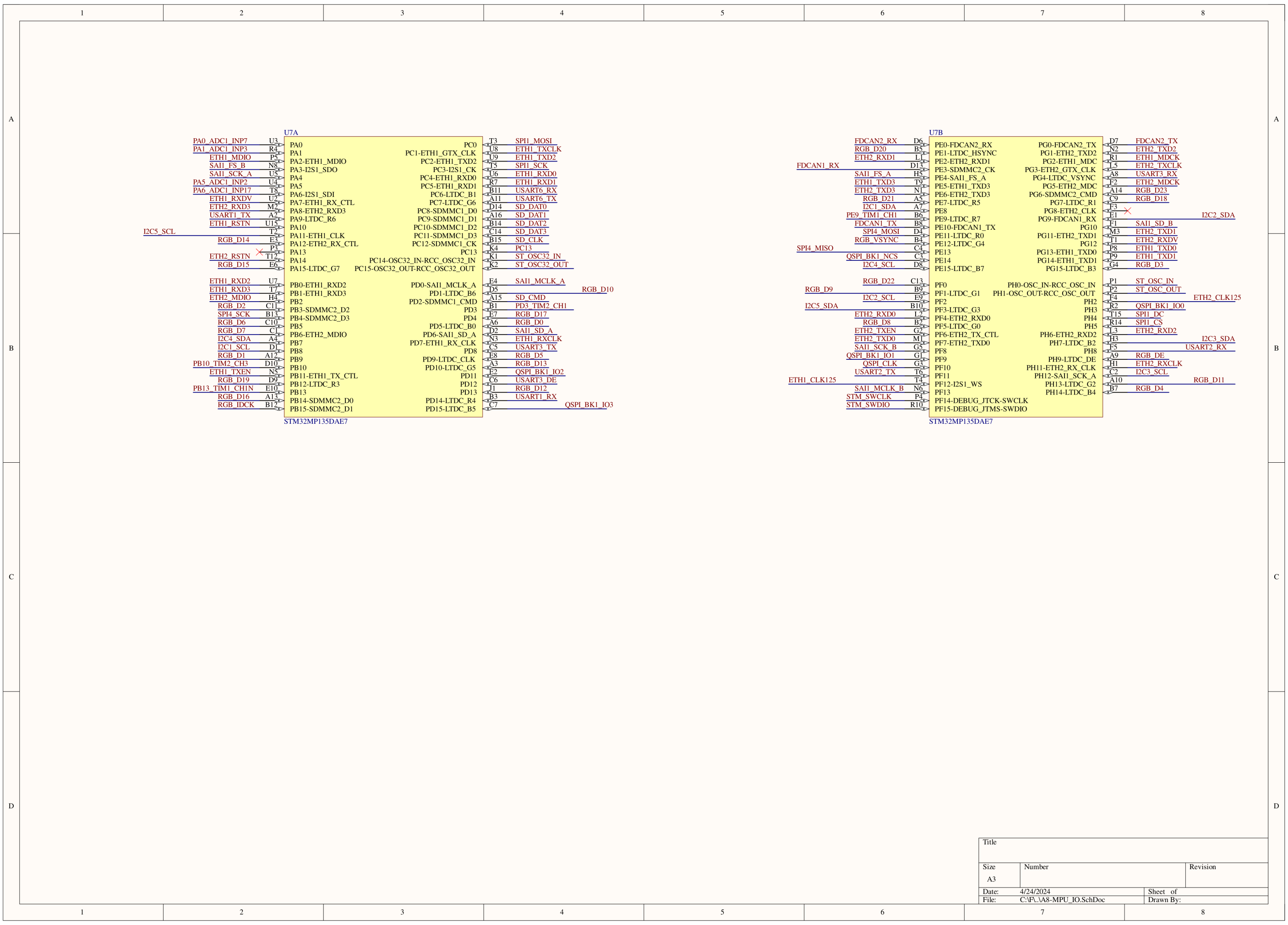
U7A
STM32MP135DAE7
U7B
STM32MP135DAE7
Title
Size
A3
Number
Revision
Date: 4/24/2024
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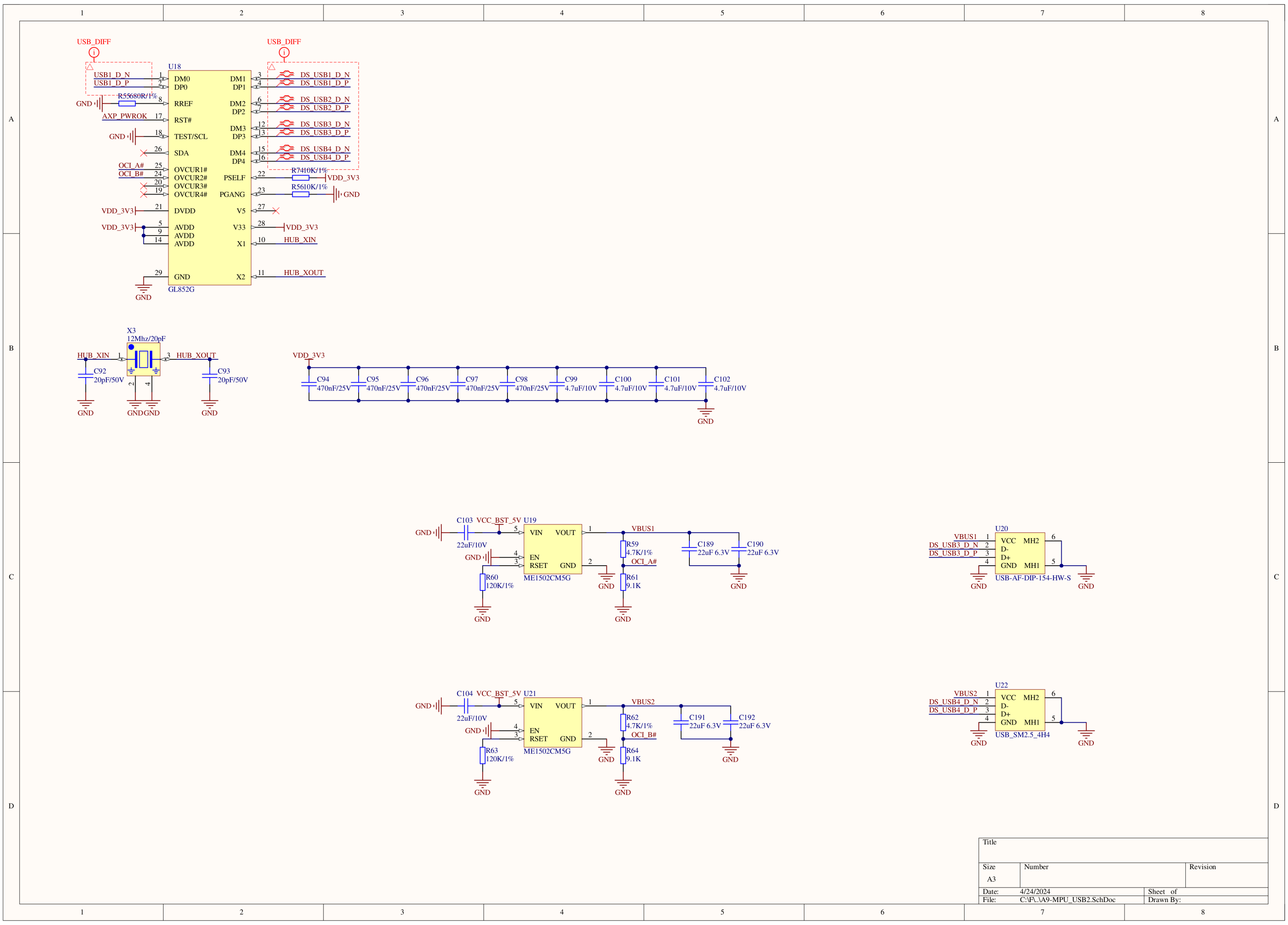
U18
GL852G
USB_DIFF
X3
12Mhz/20pF
HUB_XIN
HUB_XOUT
VDD_3V3
U19
ME1502CM5G
VBUS1
U21
VBUS2
U20
USB-AF-DIP-154-HW-S
U22
USB_SM2.5_4H4
Title
Size
A3
Number
Revision
Date: 4/24/2024
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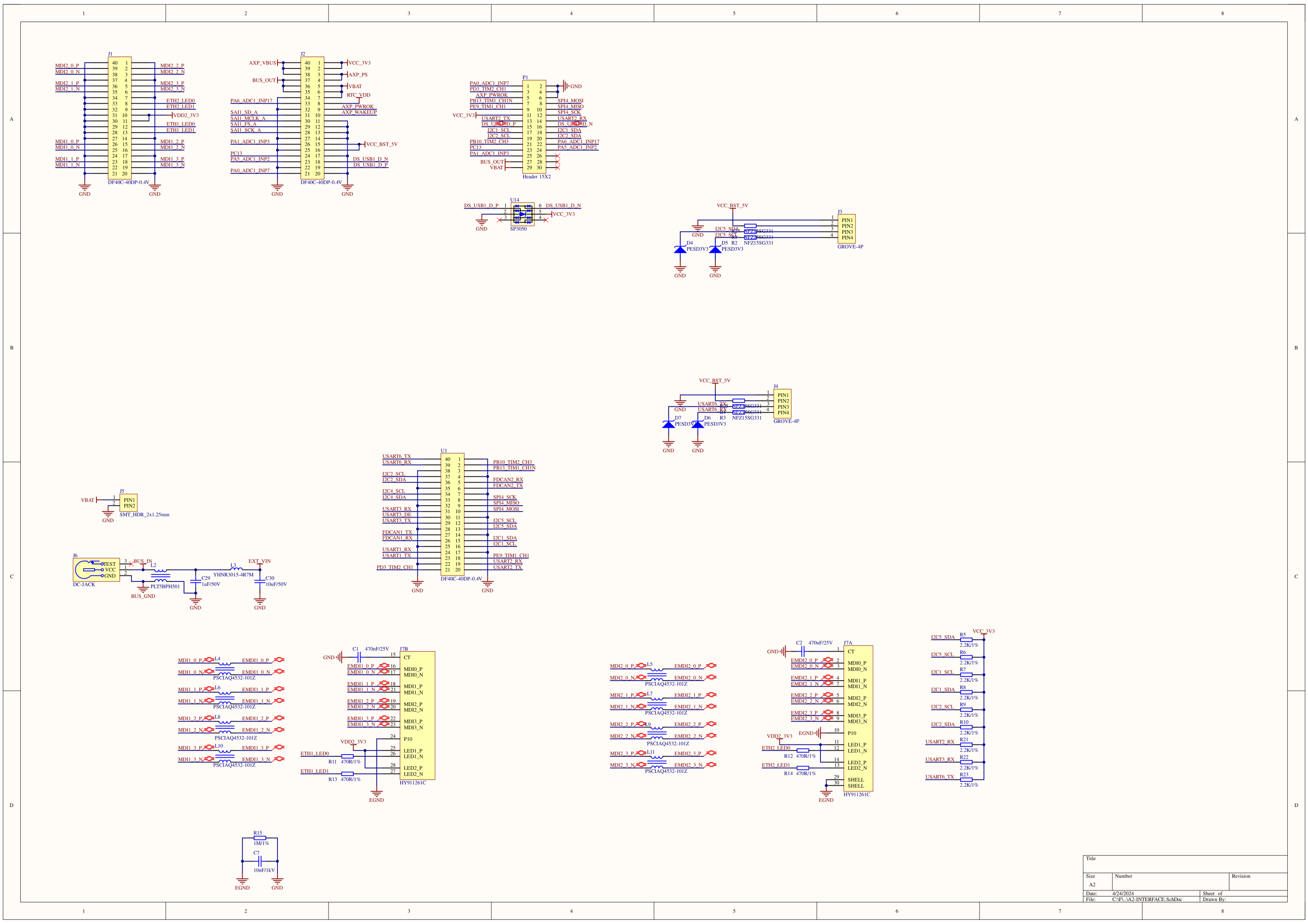

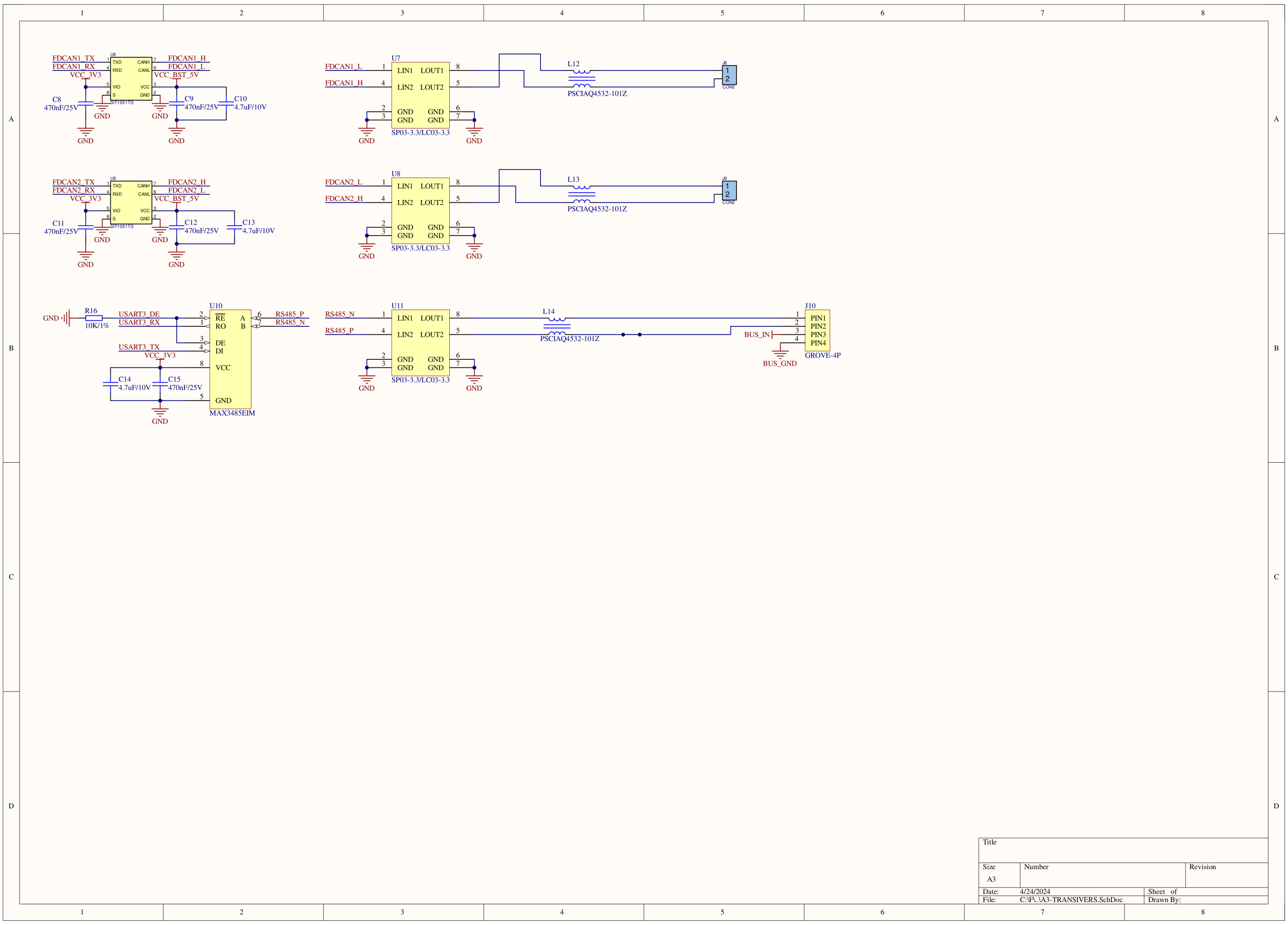

FDCAN1_TX
FDCAN1_RX
VCC_3V3
FDCAN1_H
FDCAN1_L
VCC_BST_5V
C8
470nF/25V
C9
470nF/25V
C10
4.7uF/10V
GND
U7
LIN1
LOUT1
LIN2
LOUT2
SP03-3.3/LC03-3.3
L12
PSCIAQ4532-101Z
FDCAN2_TX
FDCAN2_RX
FDCAN2_H
FDCAN2_L
C11
470nF/25V
C12
470nF/25V
C13
4.7uF/10V
U8
L13
R16
10K/1%
USART3_DE
USART3_RX
USART3_TX
U10
RE
RO
DE
DI
VCC
MAX3485EIM
RS485_P
RS485_N
C14
4.7uF/10V
C15
470nF/25V
U11
L14
J10
PIN1
PIN2
PIN3
PIN4
GROVE-4P
BUS_IN
BUS_GND
Title
Size
A3
Number
Revision
Date:
4/24/2024
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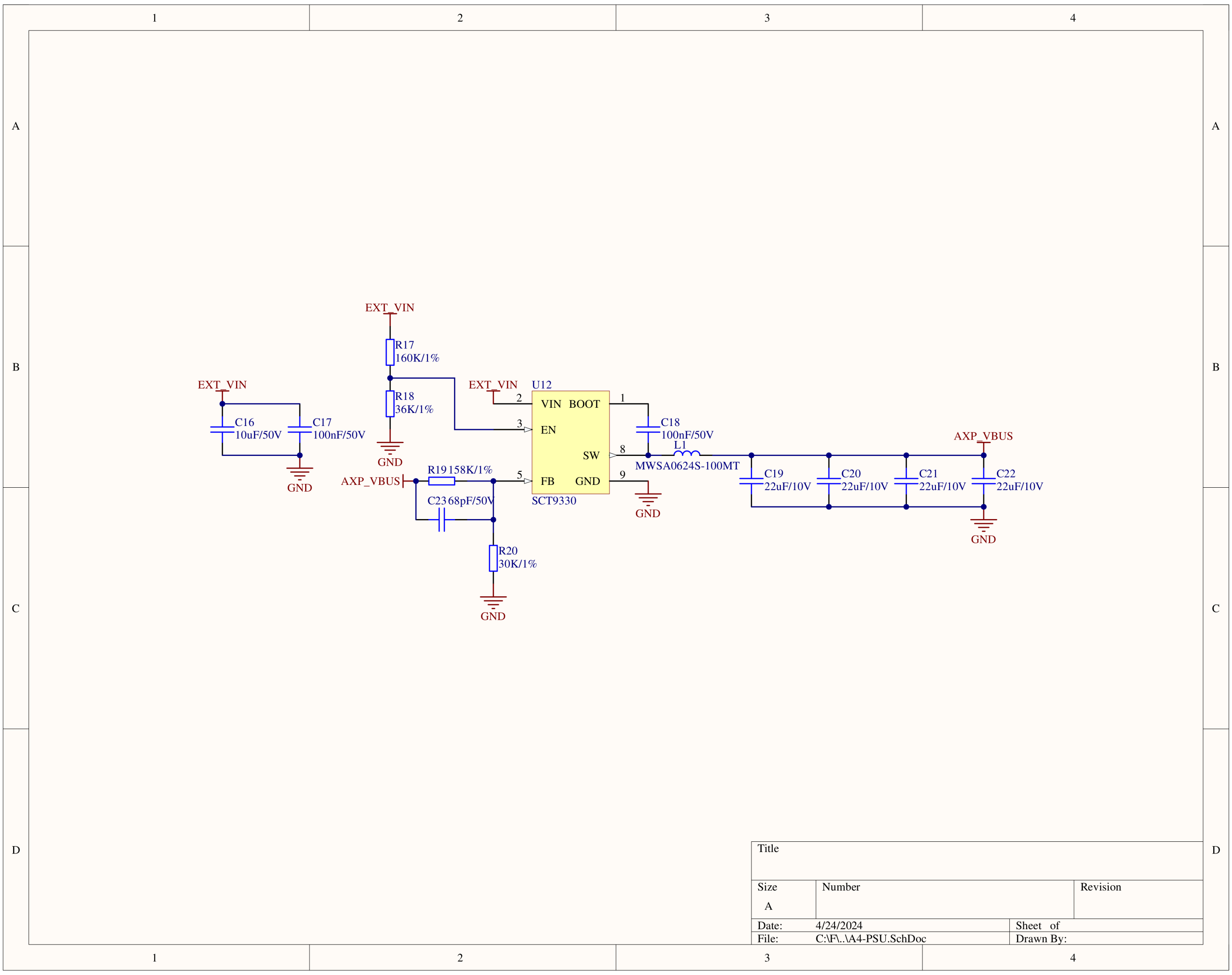

1
2
3
4
A
B
C
D
EXT_VIN
R17
160K/1%
R18
36K/1%
GND
EXT_VIN
C16
10uF/50V
C17
100nF/50V
GND
EXT_VIN
U12
VIN
BOOT
EN
SW
FB
GND
SCT9330
C18
100nF/50V
L1
MWSA0624S-100MT
GND
AXP_VBUS
R19
158K/1%
C23
68pF/50V
R20
30K/1%
GND
AXP_VBUS
C19
22uF/10V
C20
22uF/10V
C21
22uF/10V
C22
22uF/10V
GND
Title
Size
A
Number
Revision
Date:
4/24/2024
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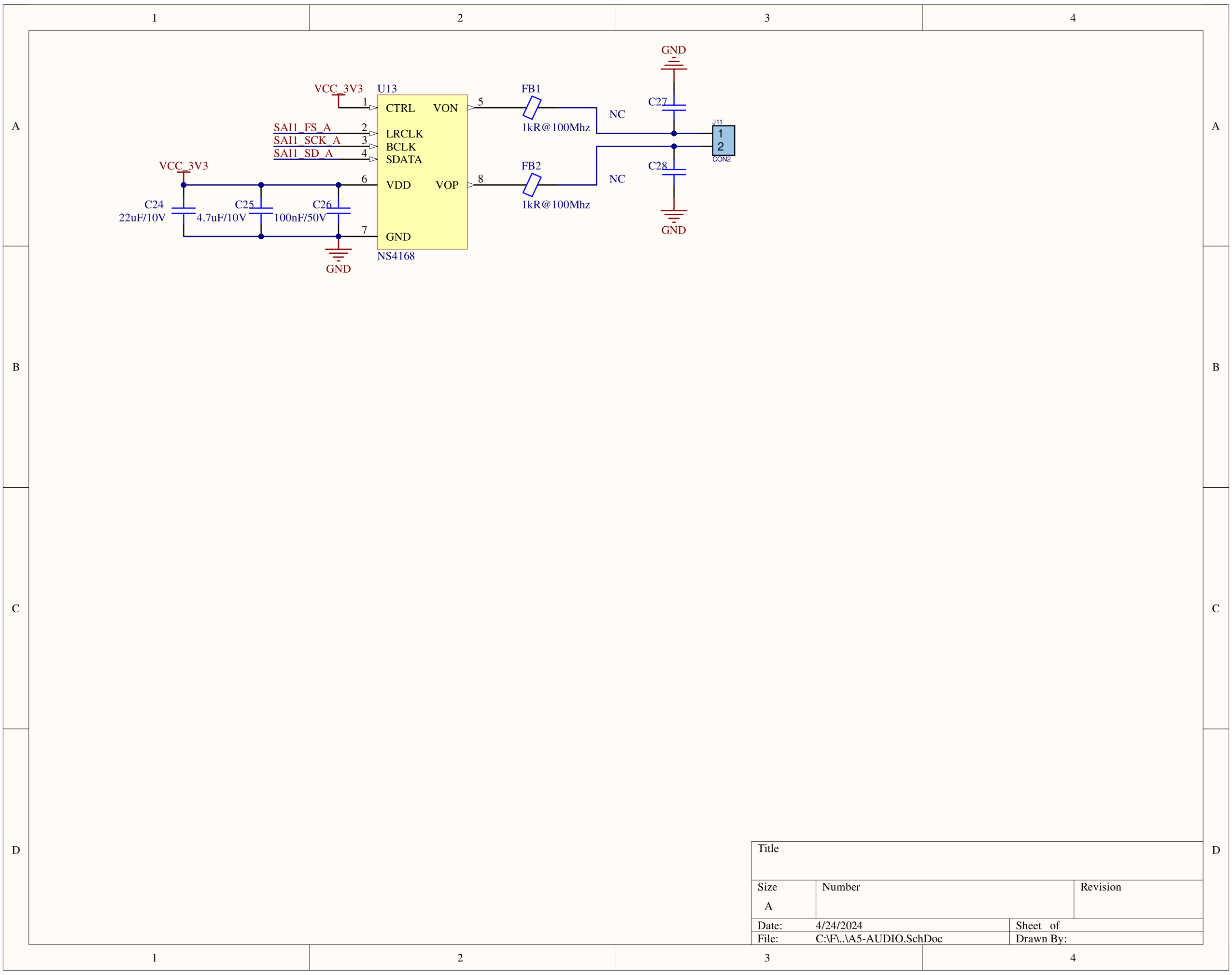
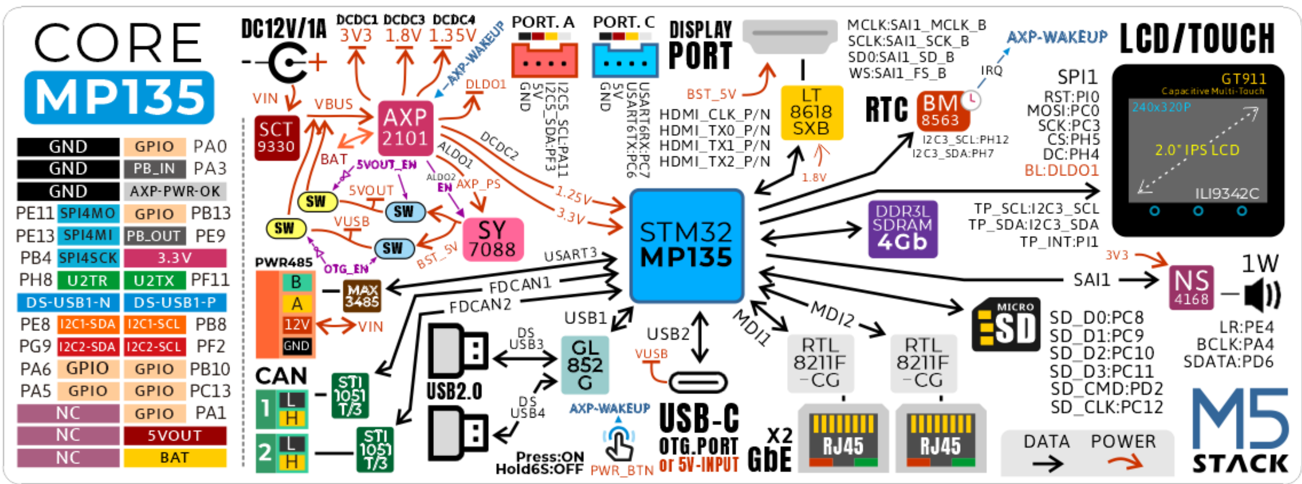


VCC_3V3
U13
CTRL
VON
SAI1_FS_A
SAI1_SCK_A
SAI1_SD_A
LRCLK
BCLK
SDATA
VDD
VOP
GND
NS4168
FB1
1kR@100Mhz
FB2
1kR@100Mhz
NC
C27
C28
GND
C24
22uF/10V
C25
4.7uF/10V
C26
100nF/50V
Title
Size
A
Number
Revision
Date:
4/24/2024
Sheet of
File:
C:\F\..\A5-AUDIO.SchDoc
Drawn By:

管脚映射



CORE
MP135
GND GPIO PA0
GND PB_IN PA3
GND AXP-PWR-OK
PE11 SPI4MO GPIO PB13
PE13 SPI4MI PB_OUT PE9
PB4 SPI4SCK 3.3V
PH8 U2TR U2TX PF11
DS-USB1-N DS-USB1-P
PE8 I2C1-SDA I2C1-SCL PB8
PG9 I2C2-SDA I2C2-SCL PF2
PA6 GPIO GPIO PB10
PA5 GPIO GPIO PC13
NC GPIO PA1
NC 5VOUT
NC BAT
DC12V/1A
DCDC1 3V3
DCDC3 1.8V
DCDC4 1.35V
AXP-WAKEUP
PORT. A
PORT. C
DISPLAY PORT
MCLK:SAI1_MCLK_B
SCLK:SAI1_SCK_B
SD0:SAI1_SD_B
WS:SAI1_FS_B
AXP-WAKEUP
LCD/TOUCH
AXP 2101
SCT 9330
SY 7088
LT 8618 SXB
RTC
BM 8563
I2C3_SCL:PH12
I2C3_SDA:PH7
HDMI_CLK_P/N
HDMI_TX0_P/N
HDMI_TX1_P/N
HDMI_TX2_P/N
SPI1
RST:PI0
MOSI:PC0
SCK:PC3
CS:PH5
DC:PH4
BL:DLDO1
GT911
240x320P
2.0" IPS LCD
ILI9342C
STM32 MP135
DDR3L SDRAM 4Gb
TP_SCL:I2C3_SCL
TP_SDA:I2C3_SDA
TP_INT:PI1
SAI1
NS 4168
1W
LR:PE4
BCLK:PA4
SDATA:PD6
PWR485
MAX 3485
FDCAN1
FDCAN2
CAN
STI 1051 T/3
USB1
USB2
GL 852 G
USB2.0
USB-C OTG.PORT or 5V-INPUT
Press:ON Hold6S:OFF
PWR_BTN
MDI1
MDI2
RTL 8211F -CG
X2 GbE
RJ45
MICRO SD
SD_D0:PC8
SD_D1:PC9
SD_D2:PC10
SD_D3:PC11
SD_CMD:PD2
SD_CLK:PC12
DATA
POWER
M5 STACK

M5-Bus

FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	PA0	GPIO
	GND	3	4	PA3	PB_IN
	GND	5	6	AXP-PWR-OK	
SPI4MO	PE11	7	8	PB13	GPIO
SPI4MI	PE13	9	10	PE9	PB_OUT
SPI4SCK	PB4	11	12	3V3	
U2RX	PH8	13	14	PF11	U2TX
	DS-USB1-N	15	16	DS-USB1-P	
I2C1-SDA	PG8	17	18	PB8	I2C1-SCL
I2C2-SDA	PG9	19	20	PF2	I2C2-SCL
GPIO	PA6	21	22	PB10	GPIO
GPIO	PA5	23	24	PC13	GPIO
	NC	25	26	PA1	GPIO
	NC	27	28	5V	
	NC	29	30	BAT	

M5-Bus	STM32MP135DAE7
U2RX	PH8
U2TX	PF11
I2C1-SDA	PE8
I2C1-SCL	PB8
I2C2-SDA	PG9
I2C2-SCL	PF2

设备名称	Linux 设备节点
USART2	/dev/ttySTM2
I2C1	/dev/i2c-2
I2C2	/dev/i2c-3

PORT.A

PORT.A	I2C5_SCL	I2C5_SDA
STM32MP135DAE7	PA11	PF3

设备名称	Linux 设备节点
UI2C5	/dev/i2c-1

PORT.C

PORT.C	USART6RX	USART6TX
STM32MP135DAE7	PC6	PC7

设备名称	Linux 设备节点
USART6	/dev/ttySTM0

RS485

MAX3485EIM	USART3RX	USART3TX	DE/RE
STM32MP135DAE7	PG4	PD8	PD12

设备名称	Linux 设备节点
USART3	/dev/ttySTM3

CAN

STM32MP135DAE7	PE3	PE10	PG0	PE0
SIT1051T/3(FDCAN1)	FDCAN1_TX	FDCAN1_RX		
SIT1051T/3(FDCAN2)			FDCAN2_TX	FDCAN2_RX

Display

LT8618SXB	MCLK	SCLK	SD0	WS	I2C3_SDA	I2C3_SCL
STM32MP135DAE7	PF13	PF8	PA3	PG10	PH7	PH12

设备名称	Linux 设备节点
I2C3	/dev/i2c-0

RTC

PORT.A	I2C3_SCL	I2C3_SDA
STM32MP135DAE7	PH7	PH12

设备名称	Linux 设备节点
I2C3	/dev/i2c-0

microSD

microSD	SD_DAT0	SD_DAT1	SD_DAT2	SD_DAT3	SD_CMD	SD_CLK
STM32MP135DAE7	PC8	PC9	PC10	PC11	PD2	PC12

NS4168

NS4168	LRCLK	BCLK	SDATA	WS	I2C3_SDA	I2C3_SCL
STM32MP135DAE7	PE4	PA4	PD6	PG10	PH7	PH12

设备名称	Linux 设备节点
I2C3	/dev/i2c-0

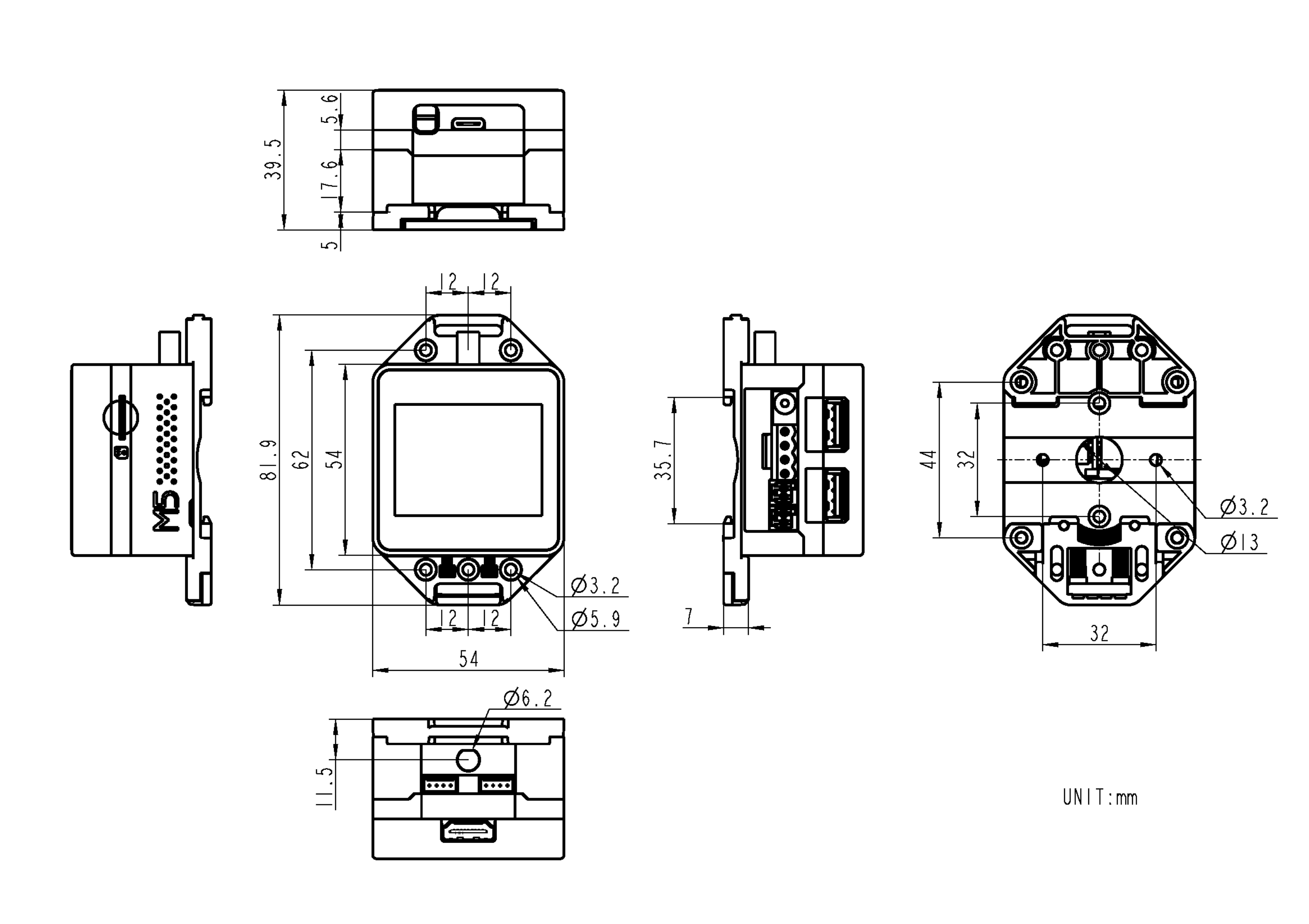
Screen&Touch

STM32MP135DAE7	ILI9342C	FT6336U
PI0	RST	
PC0	MOSI	
PC3	SCK	
PH5	CS	
PH4	DC	
PH12 (I2C3_SCL)		TP_SCL
PH7 (I2C3_SDA)		TP_SDA

AX2101	DLDO1
ILI9342C	BL

设备名称	Linux 设备节点
I2C3	/dev/i2c-0

尺寸图



数据手册

- STM32MP135DAF7
- GL852G(USB Hub Chip)
- LT8618SXB
- RTL8211F(Ethernet chip)
- FT6336U(Touch Screen Driver)
- NS4168(amplifier chip)
- BM8563(Clock Chip)
- AXP2101(PMU)
- SIT1051T/3(CAM Comunication)
- ILI9342C(Screen Driver)

软件开发

快速上手

- CoreMP135 镜像烧录与软件更新
- CoreMP135 应用开发框架
- CoreMP135 根文件系统扩容
- CoreMP135 网络配置
- CoreMP135 Buildroot 编译

- CoreMP135 覆盖设备树

UiFlow2

- CoreMP135 UiFlow2 快速上手

SDK

- M5Stack Linux Libs
- CoreMP135 Buildroot

> **Buildroot**
>
> Buildroot 是一个简单、高效、易用的嵌入式生成工具，该存储库是一个 Buildroot BR2_EXTERNAL 树，专门用于支持 STM32MP1 平台。

- CoreMP135 Buildroot External ST

镜像文件

镜像版本	内核版本	下载链接
M5_CoreMP135_buildroot_20240515	5.15.118	Download
M5_CoreMP135_buildroot_20240628	5.15.118	Download
M5_CoreMP135_debian12_20240515	5.15.118	Download
M5_CoreMP135_debian12_20240628	5.15.118	Download
M5_CoreMP135_debian12_20240919	5.15.118	Download

相关视频

- CoreMP135 简介

K135 CoreMP135 视频.mp4

- CoreMP135 镜像烧录

coremp135_image.mp4

- 基于 M5Stack Linux 应用开发框架编程控制 CoreMP135 上的外设硬件

coremp135_develop.mp4

- CoreMP135 UiFlow2 快速上手

cmp135UIFlow2quickstart.mp4